

Detailed breakdown of backtesting metrics			
Category	Metric	Definition	Significance
1. Core Metrics	Start	Start date of the backtest period.	Defines the beginning of your strategy’s testing window.
	End	End date of the backtest period.	Marks the end of your testing window.
	Duration	Total time span of the backtest (End - Start).	Indicates the robustness of the strategy across different market conditions.
	Exposure Time [%]	Percentage of time your strategy held positions (invested in the market).	A high percentage suggests frequent trading or long holding periods.
2. Returns & Profitability	Equity Final [\$]	Final portfolio value after all trades and costs.	The ultimate profit/loss of your strategy.
	Equity Peak [\$]	Highest portfolio value during the backtest.	Shows the maximum unrealized gain before drawdowns.
	Return [%]	Total return of the strategy (profit/loss as a percentage of initial capital).	Measures overall profitability.
	Buy & Hold Return [%]	Return from simply buying and holding the asset (no trading).	Benchmark to compare your strategy’s performance against passive investing.
	Return (Ann.) [%]	Annualized return (geometric mean return per year)	It smooths returns for easier comparison across strategies.
	CAGR [%]	Compound Annual Growth Rate (annualized growth rate assuming reinvestment).	Reflects consistent growth over time.

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3. Risk & Volatility	Volatility (Ann.) [%]	Annualized standard deviation of returns.	Measures risk (higher = more unpredictable returns).
	Max. Drawdown [%]	Largest peak-to-trough decline in portfolio value.	Worst-case loss; critical for risk tolerance.
	Avg. Drawdown [%]	Average of all drawdowns during the backtest.	Typical loss magnitude during downturns..
	Max. Drawdown Duration	Longest time to recover from a drawdown.	Shows resilience; shorter durations are better.
4. Risk-Adjusted Returns	Sharpe Ratio	Excess return per unit of risk (volatility). Formula: (Return - Risk-Free Rate) /Volatility.	>1 = good, >2 = excellent.
	Sortino Ratio	Similar to Sharpe, but penalizes only downside volatility.	Focuses on harmful risk; >1 is desirable.
	Calmar Ratio	Return relative to maximum drawdown. Formula: CAGR / Max Drawdown.	Higher = better recovery from worst losses.
5. Strategy Efficiency	Alpha [%]	Excess return relative to a benchmark (e.g., S&P 500).	>0 = outperforms the market.
	Beta	Sensitivity to market movements (1 = moves with the market).	<1 = less volatile than the market; >1 = more volatile.
	Profit Factor	Gross profit / Gross loss.	>1 = profitable strategy.
	Expectancy [%]	Average profit per trade (adjusted for win/loss rates).	>0 = strategy is profitable on average.

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6. Trade Statistics	# Trades	Total number of trades executed.	High frequency may increase costs (commissions).
	Win Rate [%]	Percentage of profitable trades.	High win rate ≠ profitability if losses are large.
	Best Trade [%]	Highest return from a single trade.	Highlights outlier successes.
	Worst Trade [%]	Largest loss from a single trade.	Identifies risk of catastrophic losses.
	Avg. Trade [%]	Average return per trade.	Combines win rate and risk/reward.
	Avg. Trade Duration	Average time a trade is held.	Short durations suggest scalping; long durations suggest swing trading.
7. Advanced Metrics	SQN (System Quality Number)	Measures strategy robustness. Formula: $\text{Expectancy} / \text{StdDev}(\text{Returns}) * \sqrt{\text{Trades}}$	>2 = excellent, <1 = poor.
	Kelly Criterion	Optimal position size to maximize growth. Formula: $\text{Win\%} - (\text{Loss\%} / (\text{Avg Win} / \text{Avg Loss}))$.	Suggests aggressive/conservative sizing.